Evaluation of Gridded Rainfall Products in The Bangladesh Portions of The Ganges Brahmaputra and Meghna (GBM) Basins

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1. Abstract

This project will evaluate CHIRPS, PERSIANN-CDR, CMORPH, and the CPC gauge-based product against Bangladesh Meteorological Department (BMD) rainfall observations for 2008–2023. I will analyze the Bangladesh portions of the Ganges, Brahmaputra, and Meghna separately using spatially weighted monthly rainfall, annual summaries, and statistical performance measures. A preliminary Meghna analysis has established the Python workflow and produced product rankings. I will extend this workflow to the other two basin portions and test whether rankings depend on basin, rainfall characteristic, and scoring choices. The final outputs will include a reproducible notebook, basin-specific comparisons, and recommendations for selecting rainfall inputs for subsequent hydrological studies.

2. Background And Importance

Reliable rainfall estimates support water management, agricultural assessments, and flood-related research. Uneven station coverage makes gridded products useful, but their accuracy requires regional evaluation. Montes et al. (2021) evaluated monsoon rainfall extremes against BMD observations. Related studies examined rainfall products in Meghna (Khairul et al., 2018) and inconsistent precipitation trends across GBM (Khatiwada & Curtis, 2021).

3. Research Questions And Hypothesis

The study will ask:

(1) How accurately do the products reproduce BMD monthly basin rainfall and its annual summaries?

(2) Does performance differ among the three basin portions?

(3) Do rankings change among annual totals, mean monthly rainfall, and maximum and minimum monthly rainfall?

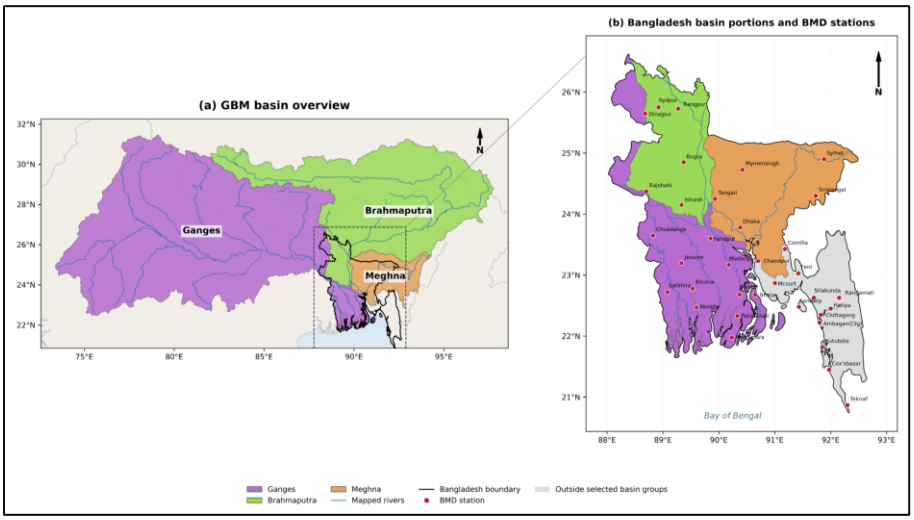
I hypothesize that performance will vary by basin and rainfall characteristic; agreement in annual totals will not necessarily indicate agreement in the wettest or driest month.

4. Data and Methods

Study Area and Datasets

The study will cover the Bangladesh portions of the Ganges, Brahmaputra, and Meghna basins during January 2008–December 2023 (Figure 1). HydroBASINS level-4 units clipped to the Bangladesh boundary define preliminary study areas of 45,878, 28,788, and 38,480 km², respectively. These map-derived areas depend on the selected basin boundaries. The provisional assignment of the southwest delta to Ganges and the treatment of shared confluence areas will be reviewed before the final analysis. BMD monthly rainfall records from 35 stations will provide the reference dataset; each basin estimate will include only stations assigned to that basin.

Figure 1. Study area showing the overall portions of the Ganges-Brahmaputra-Meghna Basins. Only the Bangladesh portions will be evaluated. The national network provides 35 station records; each basin analysis uses only stations assigned to its selected polygon. Basin assignments will be checked before the final comparison.



The comparison will include CHIRPS v2.0 monthly rainfall (0.05°), PERSIANN-CDR v01r01 daily rainfall (0.25°), bias-corrected CMORPH V1.0 daily rainfall (0.25°), and CPC Global Unified daily rainfall (0.5°). CHIRPS integrates satellite estimates and station observations (Funk et al., 2015), whereas CPC provides a gauge-based benchmark. Data preparation, spatial analysis, and statistical evaluation will be implemented in Python within a reproducible Jupyter Notebook.

Rainfall Analysis And Product Evaluation

- **Basin Rainfall Estimation:** BMD records will be checked for duplicates, missing values, inconsistent units, and station-location errors. Basin rainfall will be estimated using station weights derived from clipped Thiessen polygons. Gridded rainfall will be averaged using grid-cell overlap areas within the same boundaries. The Meghna analysis uses five stations; weights will be calculated separately for Ganges and Brahmaputra.
- **Temporal Matching:** Daily estimates will be aggregated only for complete months. Comparisons will use common valid months across BMD and all products, with annual statistics requiring 12 complete months. Missing values will remain unfilled, and excluded periods will be documented.
- **Performance Evaluation:** Products will be evaluated against BMD using RMSE, RSR, NSE, R^2, PBIAS, and KGE for monthly rainfall and annual summaries, following Mohib et al. (2025). Annual summaries will include total, mean monthly, maximum monthly, and minimum monthly rainfall. Positive PBIAS will indicate overestimation, and KGE will use the coefficient-of-variation ratio.
- **Product Ranking And Sensitivity Analysis:** An adapted multi-criteria analysis will normalize metrics to a 0–10 scale and combine them using predefined weights. Rankings will be compared across basins and rainfall

characteristics. Sensitivity tests will apply equal characteristic weights and exclude mean monthly rainfall to assess its redundancy with annual totals.

- **Uncertainty And Limitations:** Station-removal tests will assess sensitivity to gauge coverage. Interpretation will consider product resolution, missing data, accumulation periods, and potential overlap between product inputs and BMD observations. Conclusions will concern monthly and annual rainfall; daily extremes require daily observations (Zhang et al., 2011).

5. Preliminary Work Completed for Meghna Basin

The Meghna workflow is complete for the available common sample. Figure 2 shows that the products capture some shared year-to-year variation, including the high annual rainfall in 2017, but differ in magnitude. CPC and CMORPH tend to exceed BMD annual totals in several later years. CHIRPS and PERSIANN-CDR are generally closer in those years, although agreement varies by rainfall characteristic. The figure excludes 2011 and 2012, leaving 14 complete annual comparisons within the 2008–2023 window. Exact retained monthly counts will be reported from the audit files.

Figure 2. Preliminary Meghna comparison with BMD for annual total, mean monthly, maximum monthly, and minimum monthly rainfall. Maximum and minimum are calculated from monthly rainfall totals. These are completed pilot results, not findings for Ganges or Brahmaputra.

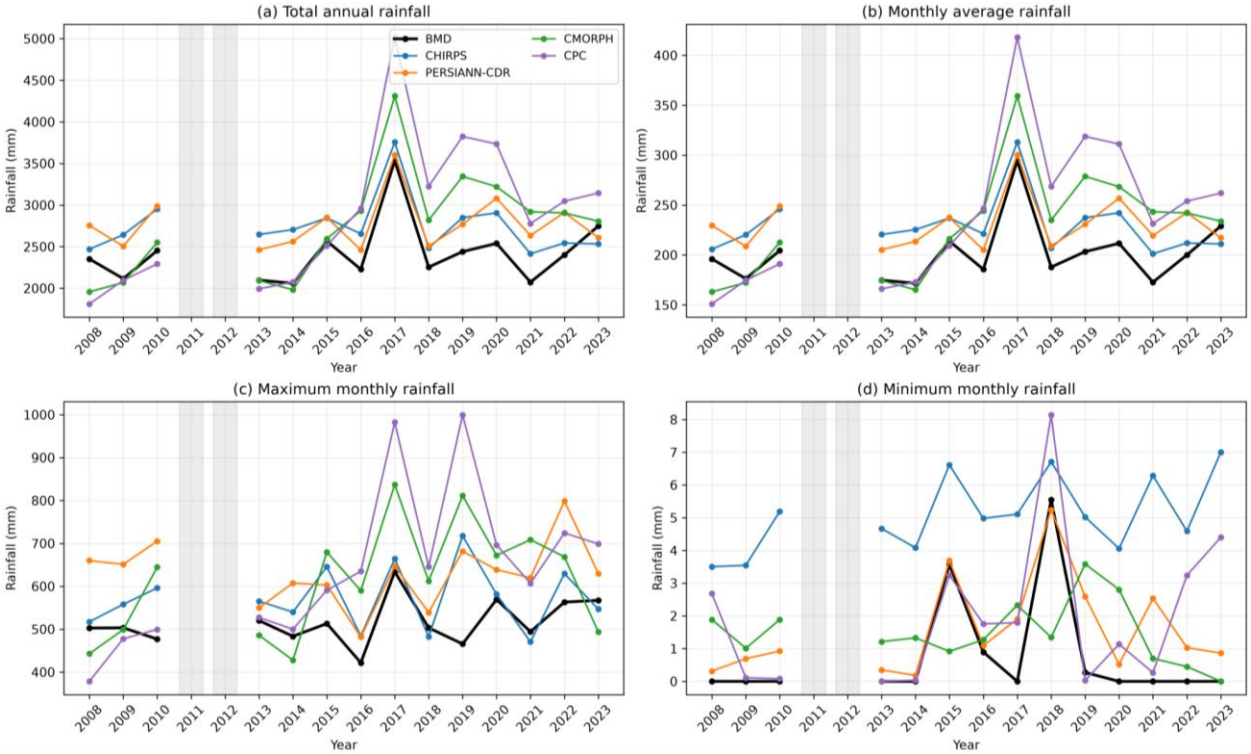


Table 1 summarizes the adapted MCA. PERSIANN-CDR ranks first overall (9.241), followed by CHIRPS (8.213), CMORPH (6.372), and CPC (1.446). CHIRPS has slightly higher total and mean contributions and a higher maximum contribution, whereas PERSIANN-CDR gains a substantially larger minimum contribution. Thus, the overall ranking depends on which rainfall characteristics receive weight; it does not show that one product is uniformly superior.

Table 1. Preliminary Meghna MCA contributions and overall ranking. Scores are dimensionless and relative to these four products; zero is the lowest normalized contribution, not zero rainfall or zero absolute skill. Rounded contributions may not sum exactly to the displayed overall score.

Product	Total	Mean	Maximum	Minimum	Overall	Rank
PERSIANN-CDR	2.871	3.828	1.047	1.495	9.241	1
CHIRPS	2.878	3.837	1.437	0.062	8.213	2
CMORPH	1.965	2.619	0.613	1.175	6.372	3
CPC	0.000	0.000	0.150	1.296	1.446	4

6. Timeline

The completed Meghna analysis provides the starting point for the four-week plan in Table 2. Approximately 10–12 hours per week will be allocated to programming and analysis, plus writing time. The schedule will follow course deadlines, with the three-basin comparison taking priority if time becomes limited.

Table 2. Work plan and expected outputs for the remaining project.

Timeline	Planned Activities	Expected Outputs
Week 1	Verify basin boundaries, BMD station locations, and data completeness. Calculate separate station weights for Ganges and Brahmaputra.	Verified station assignments and basin-specific weights.
Week 2	Extend the Meghna workflow to Ganges and Brahmaputra. Extract product rainfall, calculate monthly basin averages, and identify common valid periods.	Matched BMD and product rainfall datasets for all three basins.
Week 3	Calculate performance metrics and MCA rankings. Compare basins and test sensitivity to scoring weights and station selection.	Basin-specific performance tables, rankings, and sensitivity results.
Week 4	Finalize figures, interpret findings, document limitations, and organize the notebook and final report.	Completed report, reproducible notebook, and presentation materials.

7. Expected Outcomes and Broader Impacts

The project will identify product strengths and limitations in each basin portion and determine whether rankings change with rainfall characteristic. Outputs will include a reproducible notebook, weighted rainfall datasets, comparison figures, and basin-specific performance summaries. These findings could support rainfall-data selection for agricultural assessments, drainage studies, and hydrological modeling. Conclusions will concern historical rainfall agreement within Bangladesh, not operational flood-forecast skill or rainfall across the full transboundary GBM system.

References

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